Prodigy Anesthesia Quick Review

Trauma Surgery

Where is the proper placement of IV access in trauma patients with injuries to the superior or inferior vena cava?	proximal to the vena caval disruptions *Nagelhout JJ, Zaglaniczny KL. Nurse Anesthesia. 4th ed. Philadelphia, PA: WB Saunders Company; 2010: 881.*
What is the most common cause of hypotension in the trauma patient?	Hypovolemia. The most common causes of hypovolemia are due to disruption from major vessels of the pelvis, chest and the abdomen. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 919-921.*
How is laryngoscopy performed in all trauma patients that have confirmed or suspected c-spine injuries?	Laryngoscopy should be performed with the head in a neutral position with in-line stabilization applied. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 971.*
What are the three stages of shock?	1. Compensated shock or nonprogressive shock 2. Progressive shock 3. Irreversible shock *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 920.*
What is the most common cause of coagulopathy in the trauma patient?	dilutional thrombocytopenia *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1352.*
What is a common complication seen following long-bone fractures?	Hypoxic respiratory failure. This results from microembolization of marrow fat into the venous circulation. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 927.*
What is a common complication seen with patients presenting for major trauma surgery?	Hypothermia. Open mesentery increases heat loss. Shock and anesthesia also reduce heat production. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1359.*
What induction agent is contraindicated in a trauma patient with a possible head injury?	Ketamine increases ICP and should not be used in this patient population. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 924.*
Is nitrous oxide safe to use in acute trauma case patients?	Nitrous oxide should be avoided in the acute trauma patient. Nitrous oxide diffuses into closed spaces making it contraindicated in patients with bowel injury, pneumothorax, and patients with closed-head injuries. These conditions are difficult to rule out in the acute setting and should be avoided. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 924.*

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A trauma patient presents with distended neck veins, unilateral decreased breath sounds, hypotension despite vigorous fluid resuscitation, subcutaneous emphysema of the chest, diminished chest-wall motion, and hyperresonance to percussion of one hemithorax. What is the most probable diagnosis?

Tension pneumothorax. A tension pneumothorax is a life threatening condition that creates a one-way valve of air flow into the pleural cavity. Intrapleural pressure is increased with every breath as more air is trapped within this space. Positive-pressure ventilation increases the size of a pneumothorax

Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 916.

A trauma patient presents with a possible cribriform plate injury. Is a nasal intubation an acceptible technique for securing the airway?

No. The endotracheal tube can enter the brain vault during intubation, therefore nasal intubation is contraindicated.

Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 971-972.

A patient with severe traumatic injuries is suspected of also having ischemic brain damage. Which fluid would be the least appropriate for fluid resuscitation?

In instances of possible ischemic brain damage, solutions containing dextrose should be avoided as they could exacerbate the ischemia. The only exception to this rule is if hypoglycemia is documented.

Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 921.

An awake, but intoxicated patient suffered a mid-facial fracture as a result of a head-on motor vehicle accident. What intubation technique would be contraindicated after induction of this patient?

Nasal intubation is contraindicated in this case because of the history of a mid-face fracture. It is also contraindicated in patients with basilar skull fractures.

Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 570.

A patient who has suffered major trauma exhibits changes in his phonation during the interview. You know that this may indicate an increased risk for:

A patient that exhibits changes in phonation may be suffering from damage to the trachea and vocal structures. The mechanism of injury may place the patient at risk for neurological damage during airway instrumentation, resulting in the need for a difficult airway plan to avoid damage to the cervical spine. Changes in voice may also indicate recurrent laryngeal nerve damage which can increase the risk of aspiration.

Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 736.

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